21.1: The Human Genome Project fostered development of faster, less expensive sequencing techniques

Upon sequencing the genome of any species, scientists can study whole sets of genes and their interactions, an approach called ****genomics**. The sequencing efforts that feed this approach generate enormous volumes of data. The necessity of dealing with this ever-increasing flood of information has energized the field of **bioinformatics**, the application of computational methods to store and analyze biological data.

The sequencing of the human genome, an ambitious undertaking, officially began as the **Human Genome Project** in 1990. Organized by an international, publicly funded consortium of scientists at universities and research institutes, the project involved 20 large sequencing centers in six countries plus a host of other labs working on smaller parts of the project.

After the human genome sequence was largely completed in 2003, the sequence of each chromosome was analyzed and described in a series of papers, the last of which covered chromosome 1 and was published in 2006. The sequence is over 99% complete—the figure is not 100% because numerous gaps remain, usually in regions with much repetitive DNA that are difficult to sequence. The sequenced DNA was pooled from a few individuals; scientists reviewed the results and agreed on a **reference genome**, a full sequence that best represents the genome of a species. The human reference genome is continuously being revised and re-released.

The ultimate goal in mapping any genome is to determine the complete nucleotide sequence of each chromosome. For the human genome, this was accomplished by scientists using sequencing machines and the dideoxy chain termination method mentioned in Concept 20.1. Two approaches complemented each other in this endeavor. The initial approach was a methodical one that ordered each fragment based on earlier genetic mapping of the human genome. In 1998, however, molecular biologist J. Craig Venter set up a company (Celera Genomics) and declared his intention to sequence the entire human genome using an alternative strategy. The **whole-genome shotgun approach** starts with the cloning and sequencing of DNA fragments from randomly cut DNA. Powerful computer programs then assemble the resulting very large number of overlapping short sequences into a single continuous sequence **(Figure 21.2)**. The whole-genome shotgun approach is widely used today.

Figure 21.2

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Whole-genome shotgun approach to sequencing.
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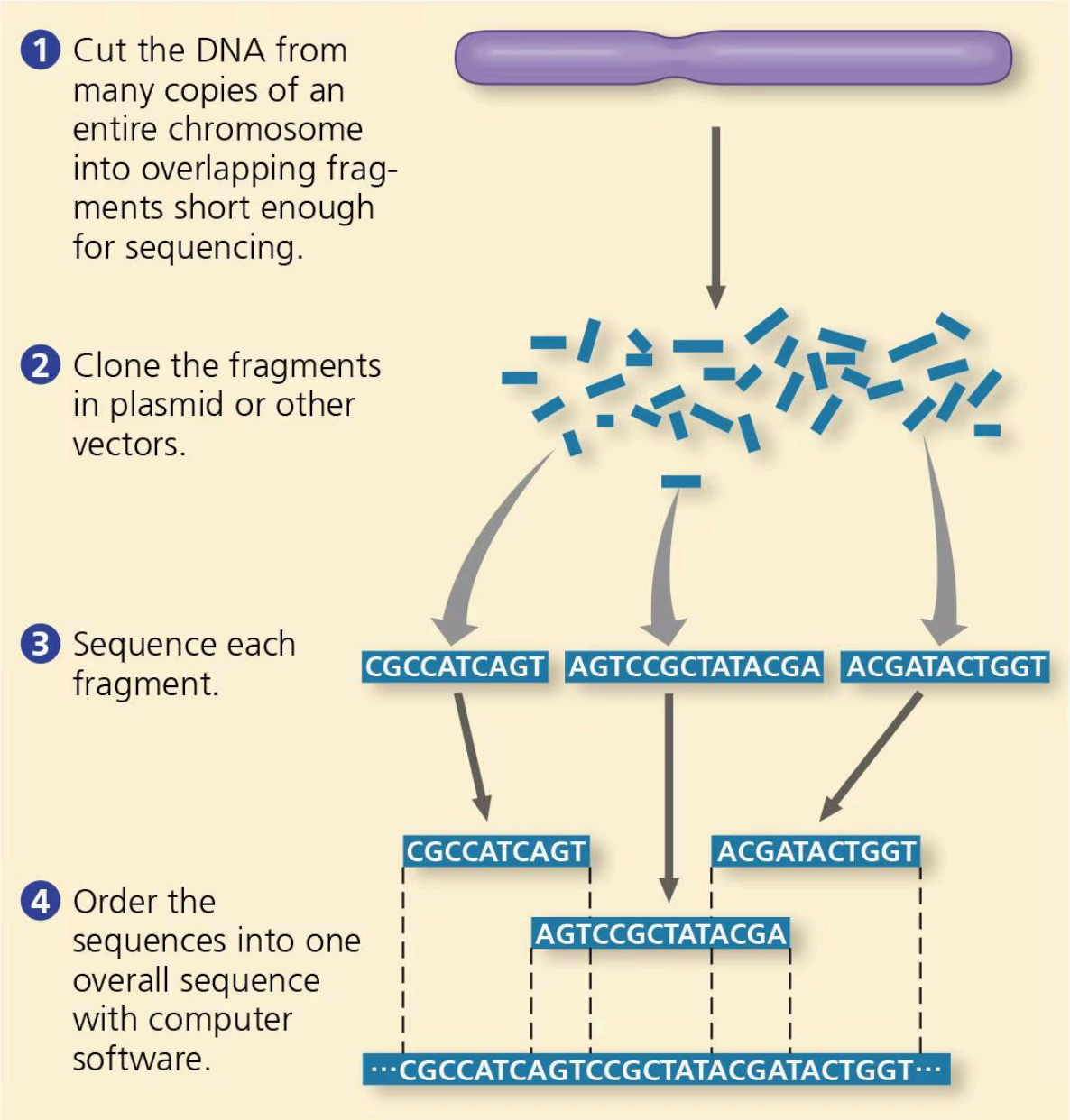

Steps are as follows.

Step 1: Cut the D N A from many copies of an entire chromosome into overlapping fragments short enough for sequencing.

Step 2: Clone the fragments in plasmid or other vectors.

Step 3: Sequence each fragment. Sample nucleotide base sequences for some fragments shown.

Step 4: Order the sequences into one overall sequence with computer software. Fragments are shown put together into one long nucleotide base sequence.

In this approach, random DNA fragments are cloned (see Figure 20.4), sequenced, and then ordered relative to each other.

VISUAL SKILLS *The fragments in step 2 of this figure are depicted as scattered, rather than arranged in an ordered array. How does this depiction reflect the approach?*

*For suggested answer, see**Appendix A*.

HHMI BioInteractive Animation

Shotgun Sequencing

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A major thrust of the Human Genome Project was the development of technology for faster sequencing (see Concept 20.1). Improvements over the years chipped away at each time-consuming step, enabling the rate of sequencing to accelerate impressively: Whereas a productive lab could typically sequence 1,000 base pairs a day in the 1980s, by the year 2000 each research center working on the Human Genome Project was sequencing 1,000 base pairs *per second*. Currently, the most widely used automated "next-generation" sequencing machines can sequence nearly 35 million base pairs per second (see Figure 20.3). Also, because of the sensitivity of these techniques, the DNA can be sequenced directly, either in fragments or as a single whole molecule; the cloning step (**STEP 2** in Figure 21.2) is unnecessary. Methods that can analyze biological materials very rapidly and produce enormous volumes of data are said to be "high-throughput." Sequencing machines using rapid sequencing techniques are an example of high-throughput devices.

Along with massive increases in sequencing speed, the cost of sequencing entire genomes has plummeted. Whereas sequencing the first human genome took 13 years and cost between $500 million and $1 billion, by 2007 the same task took four months at a cost of $1 million. In 2019, the fastest machines are able to sequence the complete genomes of 48 individuals in 44 hours for under $1,000 per genome.

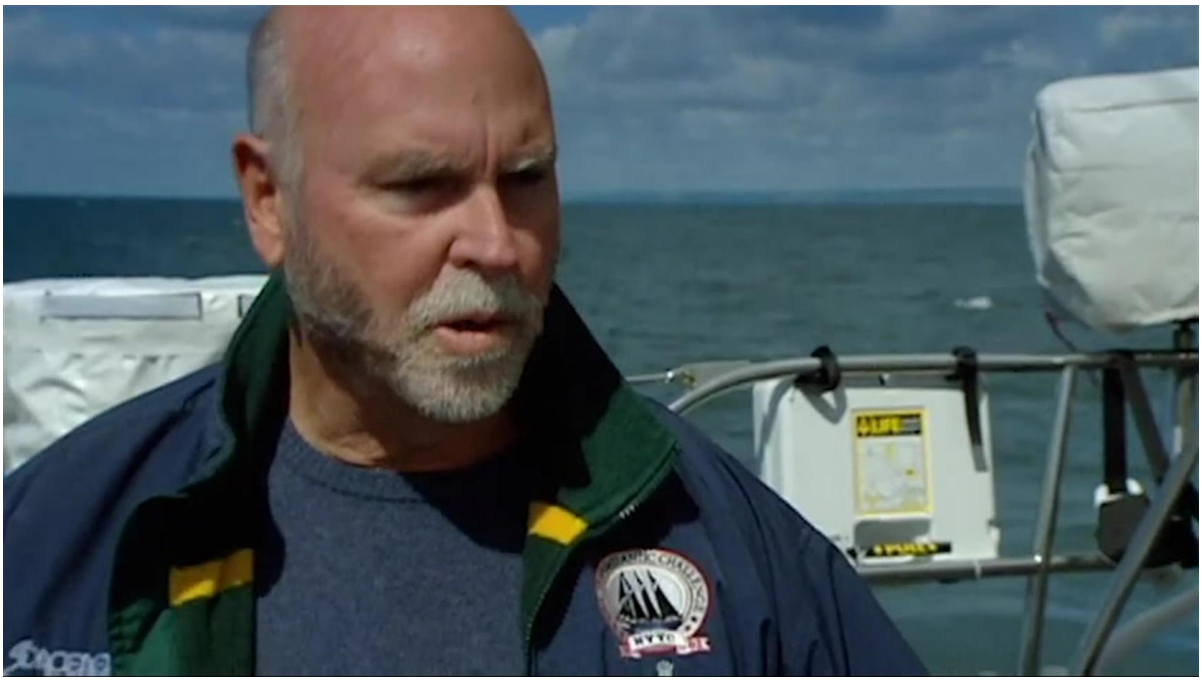
These technological advances have also facilitated an approach called **metagenomics** (from the Greek *meta*, beyond), in which DNA from an entire community of species (a *metagenome*) is collected from an environmental sample and sequenced. Again, computer software sorts out the partial sequences and

assembles them into the individual species' partial of complete genomes. An advantage of this technique is the ability to sequence the DNA of mixed microbial populations, which eliminates the need to culture each species separately in the lab, a challenge that has limited the study of microorganisms. So far, this approach has been applied to communities found in environments as diverse as the human intestine and extreme habitats like thermal springs where the temperature exceeds 80°C.

At first glance, genome sequences of humans and other organisms are simply dry lists of nucleotide bases—millions of A's, T's, C's, and G's in mind-numbing succession. Making sense of this massive amount of data has called for new analytical approaches, which we discuss next.

BBC Video

Collecting New Genes to Create New Life



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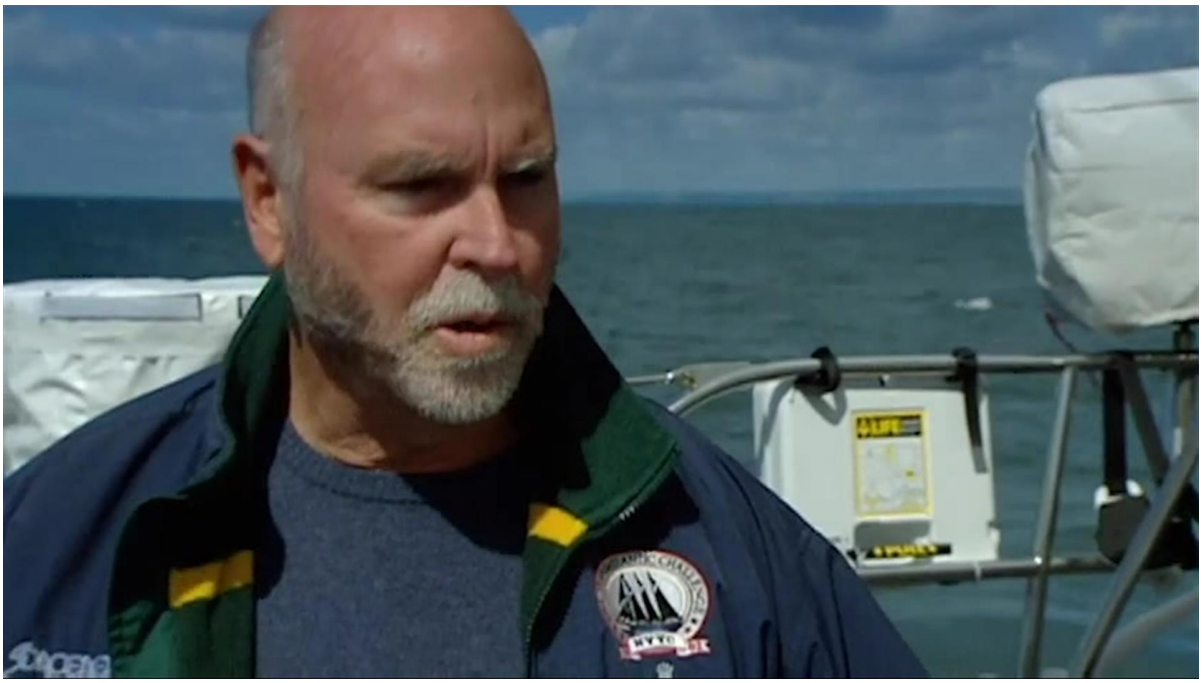

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